OurMocks

2027 CYCLE

MOCK · PRACTICE · CRACK

CUET UG

Mathematics

Algebra

Matrices, Determinants and the Matrix Method · Section A 1 + B1 II + B2 II · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Matrices — concept, notation, types	"Concept, notation, order, equality, types of matrices, zero matrix, transpose of a matrix, symmetric and skew symmetric matrices." (Section A Unit 1; Section B1 Mathematics Unit II)	Naming a matrix's type on sight; building matrices from conditions like $a_{ij} = i + j$; extracting unknowns from equality.
Operations on matrices	"Operations on matrices: Addition, multiplication and multiplication with a scalar. Simple properties of addition, multiplication and scalar multiplication. Non-commutativity of multiplication of matrices and existence of non-zero matrices whose product is the zero matrix (restrict to square matrices of order 2). Invertible matrices and proof of the uniqueness of inverse, if it exists." (Section B1 Mathematics Unit II)	Computing products fast and safely; spotting AB = BA or AB = O cases; using transpose laws without expanding.
Determinants, adjoint and inverse	"Determinant of a square matrix (up to 3 × 3 matrices), minors, cofactors and applications of determinants in finding the area of a triangle. Adjoint and inverse of a square matrix." (Section B1 Mathematics Unit II; Section B2 Applied Mathematics Unit II)	Evaluating 2 × 2 and 3 × 3 determinants; triangle-area and collinearity checks; two-step inverse construction.
Solving linear systems by the matrix method	"Consistency, inconsistency and number of solutions of system of linear equations by examples, solving system of linear equations in two or three variables (having unique solution) using inverse of a matrix." (Section B1 Mathematics Unit II)	Writing AX = B from a worded system; deciding unique / no / infinite solutions from $\|A\|$; finishing with $X = A^{-1}B$.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The official 2026 syllabus for Mathematics/ Applied Mathematics publishes this unit as **Section A Unit 1 (compulsory for every**

candidate) and again as **Section B1 Unit II** and **Section B2 Unit II** — whichever section you sit, matrices and determinants are on the table, so this chapter pays everywhere.

What a Matrix Is — and the Types You Must Name

A matrix is just a **rectangular grid of numbers** — the mathematics version of a class seating chart: rows run across, columns run down. We write a matrix as $A = [a_{ij}]_{m\times n}$, where the **order** $m \times n$ says m rows and n columns, and the element a_{ij} sits in row i, column j. Two matrices are **equal** only when they have the same order **and** every matching pair of entries is equal — which is exactly how "find x and y" questions are built. Nothing about a matrix is scary: it is a table with rules.

Type	Definition	Tiny example or test
Row matrix	Exactly one row, any number of columns.	$[1 \quad 4 \quad 9]_{\{1\times 3\}}$
Column matrix	Exactly one column.	$[2;5]_{\{2\times 1\}}$
Square matrix	Rows = columns; only these have **determinants** and inverses.	$2\times 2, 3\times 3$
Zero matrix O	Every entry is 0; the additive identity.	$A + O = A$
Diagonal matrix	Square; non-zero entries only on the main diagonal.	$[3, 0; 0, 7]$
Scalar matrix	Diagonal with **equal** diagonal entries k.	$[k, 0; 0, k]$
Identity I	Scalar matrix with $k = 1$; the multiplication identity.	$AI = IA = A$
Symmetric	Square with $A' = A$ — mirror image across the diagonal.	$a_{ij} = a_{ji}$ must hold
Skew-symmetric	Square with $A' = -A$; forces **every diagonal entry to be 0**.	$a_{ii} = 0$ for all i

Three type-facts carry a surprising number of MCQs, so pin them down now. First, the **transpose** A' is made by turning rows into columns — entry a_{ij} moves to position a_{ji}. Second, a skew-symmetric matrix can never have a non-zero diagonal, because $a_{ii} = -a_{ii}$ forces $a_{ii} = 0$. Third, "diagonal" and "scalar" and "identity" form a nesting: every identity matrix is a scalar matrix, and every scalar matrix is a diagonal matrix, but none of it runs backwards.

EVERYDAY EXAMPLE — THE EXAM SEATING CHART

Think of the exam hall chart: rows are benches, columns are the window side to the door side. The **order** "4×5" says 4 benches of 5 students. **Equality** means two schools' charts match

bench-for-bench, seat-for-seat — if even one student sits elsewhere, the charts are different. The **transpose** is the same hall viewed from the invigilator's podium, where columns become rows. And a **symmetric** chart is a hall where swapping "bench i, seat j" with "bench j, seat i" changes nothing — the picture is its own mirror image.

WORKED EXAMPLE — READING OFF UNKNOWNS FROM EQUALITY

If $[x-y, 4; 2, x+y] = [3, 4; 2, 9]$, find x and y.

Solution. Equality matches entries position by position. Top-left: $x-y=3$. Bottom-right: $x+y=9$. Adding the two equations kills y: $2x=12$, so $x=6$; back-substituting, $y=3$. Always use **adding or subtracting the pair** — it is one line, while guessing wastes time. Check: $x-y=6-3=3$ and $x+y=9$. Both match, so the pair $(6,3)$ is the answer.

Matrix Arithmetic — What Works, What Breaks

Matrix algebra looks like school algebra but **breaks in exactly three places**, and the exam tests those breaks, not the parts that behave. Addition and subtraction are harmless: they need the **same order** and run entry by entry, so $[1,2;3,4]+[0,5;6,1]=[1,7;9,5]$. Multiplying a matrix by a scalar k is equally tame — every entry gets multiplied, and $k(A+B)=kA+kB$ always. The moment you reach matrix × matrix, the rules you trust start snapping.

The product AB is built from **row-by-column dot products**: entry (i,j) of AB is row i of A dotted into column j of B. This is why the shapes must chain — $(m\times n)(n\times p)=m\times p$ — and why AB usually does not equal BA for three separate reasons: the shapes may not even both exist, the dot products pair different rows with different columns, and the results can have different sizes. **Compute honestly**, never by pattern:

THE 2 × 2 PRODUCT, ONCE AND FOR ALL

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} e & f \\ g & h \end{pmatrix}=\begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}$$

Row 1 · Column 1 → top-left entry, Row i · Column j → entry (i, j)

- **Multiplication is not commutative.** $AB=BA$ holds only in special cases (one factor scalar, or both diagonal); treat it as false until proven for the given matrices.
- **Zero divisors exist.** You **can** have $AB=O$ with $A=O$ false and $B=O$ false: take $A=[1,1;2,2]$, $B=[1,1;-1,-1]$ — every dot product collapses to $1-1=0$.
- **No cancelling.** $AB=AC$ does **not** let you cancel A; multiply A^{-1} in front of both sides only when A is invertible.
- **Powers need square matrices.** $A^2=AA$ only makes sense when rows match columns — one more reason "square" is special.

The transpose has its own law set, and one of them is famous for being **backwards**: $(AB)'=B'A'$. The order flips because transposing stacks rows into columns in reverse sequence. The full kit: $(A')'=A$, $(A+B)'=A'+B'$, $(kA)'=kA'$, $(AB)'=B'A'$, and $|A'|=|A|$. From symmetric

and skew-symmetric pieces you can rebuild **any** square matrix: $A = P + Q$ where $P = \frac{1}{2}(A + A')$ is symmetric and $Q = \frac{1}{2}(A - A')$ is skew-symmetric — a clean decomposition the MCQs love to restate in disguise.

WORKED EXAMPLE — PRODUCT, TRANSPOSE LAW, AND THE SYM + SKEW SPLIT

Let $A = [2, 1; 0, 3]$ and $B = [1, 4; 2, 0]$. Find AB, verify $(AB)' = B'A'$, and split A into symmetric + skew parts.

Solution. Dot products: top-left $2 \cdot 1 + 1 \cdot 2 = 4$; top-right $2 \cdot 4 + 1 \cdot 0 = 8$; bottom-left $0 \cdot 1 + 3 \cdot 2 = 6$; bottom-right $0 \cdot 4 + 3 \cdot 0 = 0$. So

$$AB = [4, 8; 6, 0].$$

Now $(AB)' = [4, 6; 8, 0]$. Check the flip law: $B' = [1, 2; 4, 0]$, $A' = [2, 0; 1, 3]$, and

$$B'A' = [1 \cdot 2 + 2 \cdot 1, 1 \cdot 0 + 2 \cdot 3; 4 \cdot 2 + 0 \cdot 1, 4 \cdot 0 + 0 \cdot 3] = [4, 6; 8, 0].$$

Identical — the reversed-order law holds. For the split:

$$P = \frac{1}{2}(A + A') = \left[2, \frac{1}{2}; \frac{1}{2}, 3\right] \quad \text{(symmetric, since } P' = P\text{)},$$

$$Q = \frac{1}{2}(A - A') = \left[0, \frac{1}{2}; -\frac{1}{2}, 0\right] \quad \text{(skew, since } Q' = -Q\text{)}.$$

And indeed $P + Q = [2, 1; 0, 3] = A$. Two minutes of honest arithmetic, three exam facts confirmed in one go.

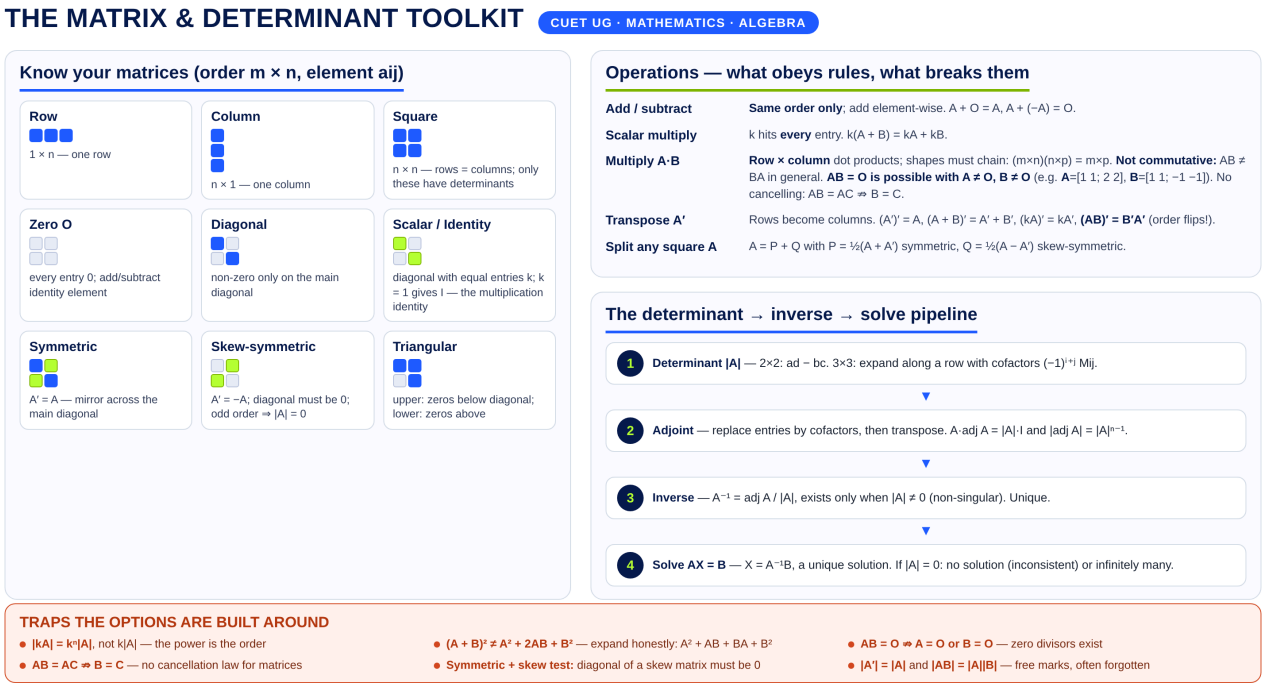


Figure 1: The full toolkit on one board: matrix types, the operation rulebook, and the determinant-to-solution pipeline every question rides on.

Determinants — The Number Inside Every Square Matrix

Every **square** matrix carries a single number called its determinant, written $|A|$ or Δ — think of it as the matrix's **fingerprint**: one value that decides whether the matrix can be inverted, whether a system of equations has a unique answer, and whether three points sit on one line. For a 2×2 matrix the computation is one crossing multiply. For 3×3 you expand along a row: each entry is multiplied by its **cofactor**, the minor M_{ij} (the determinant left after deleting row i and column j) carrying a sign $(-1)^{i+j}$. The checkerboard of signs $[+,-,+;-,+,-;+,-,+]$ never changes, so memorise it once and expand along whichever row or column has the most zeros.

DETERMINANTS — THE TWO SIZES YOU NEED

$$2 \times 2: \quad \left|\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right| = ad - bc$$

$$3 \times 3: \quad |A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}, \quad C_{ij} = (-1)^{i+j}M_{ij}$$

$$\text{Area of a triangle} = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

- **Row swap flips the sign.** Two row swaps return the original — the foundation of many MCQs.
- **Two identical rows (or columns) make the determinant 0.** A special case: a row of all zeros also gives 0.
- **Multiplying one row by k multiplies $|A|$ by k.** So multiplying **every** entry of an $n \times n$ matrix — that is, the matrix kA — multiplies the determinant n times: $|kA| = k^n|A|$.
- **Transpose leaves it alone:** $|A'| = |A|$. **Product splits:** $|AB| = |A||B|$.
- **Triangular shortcut:** for upper or lower triangular matrices, $|A|$ is just the product of the diagonal entries.

The triangle-area formula above is the determinant's most likeable application — and its sign behaviour is the testable part. The expression inside the absolute value is exactly the determinant of the matrix with rows $[x_1, y_1, 1]$, $[x_2, y_2, 1]$, $[x_3, y_3, 1]$, so when the three points are **collinear** that determinant is 0 and the "triangle" is a degenerate line with no area. Area takes the absolute value; collinearity is simply the zero test. Two question styles, one computation.

WORKED EXAMPLE — A 3 × 3 DETERMINANT, THEN AN AREA AND A COLLINEARITY CHECK

Part 1: evaluate $\left|\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{pmatrix}\right|$.

Solution. Expand along row 1:

$$= 1\left|\begin{pmatrix} 5 & 6 \\ 8 & 10 \end{pmatrix}\right| - 2\left|\begin{pmatrix} 4 & 6 \\ 7 & 10 \end{pmatrix}\right| + 3\left|\begin{pmatrix} 4 & 5 \\ 7 & 8 \end{pmatrix}\right|$$

$$= 1(50 - 48) - 2(40 - 42) + 3(32 - 35) = 2 + 4 - 9 = -3.$$

The determinant is -3, so this matrix is non-singular and invertible — we will need exactly this matrix again in the cheatsheet checks.

Part 2: find the area of the triangle with vertices $(1,2)$, $(3,-2)$, $(5,4)$; then test $(0,0)$, $(2,4)$, $(4,8)$ for collinearity.

Solution. Area $= \frac{1}{2}|1(-2-4)+3(4-2)+5(2-(-2))| = \frac{1}{2}|-6+6+20| = 10$ square units. For the collinearity test: $\frac{1}{2}|0(4-8)+2(8-0)+4(0-4)| = \frac{1}{2}|16-16| = 0$ — the three points lie on one straight line, so no triangle exists.

Adjoint, Inverse and the Matrix Method

A matrix's inverse plays the role that division plays for numbers — with one honest caveat: only **square, non-singular** matrices have one. The bridge is the adjoint: replace every entry by its cofactor, then transpose the result. Multiplying the two always lands on a very tidy identity, and everything else in this section is that identity worn in different clothes:

THE FOUR IDENTITIES THAT RUN THIS CHAPTER

$$A(\text{adj } A) = |A|I \quad (n \times n)$$

$$|\text{adj } A| = |A|^{n-1}$$

$$A^{-1} = \frac{\text{adj } A}{|A|} \quad \text{exists only when } |A| \neq 0$$

$$(AB)^{-1} = B^{-1}A^{-1} \quad \text{(order flips, like the transpose law)}$$

- **Singularity is a gate.** If $|A| = 0$, the matrix is **singular**: no inverse, no division, no matrix-method solution. If $|A| \neq 0$, it is **invertible** (non-singular).
- **The inverse is unique.** If B and C both work as inverses of A, then $B = BI = B(AC) = (BA)C = IC = C$. One line, and the syllabus explicitly names this proof — it is a gift.
- **The 2×2 shortcut.** Swap the two diagonal entries, change the sign of the other two, divide every entry by $|A|$. Write it once and never expand a cofactor for a 2×2 again.
- **Skew-symmetric matrices of odd order are always singular.** Since $A' = -A$, we get $|A| = |A'| = |-A| = (-1)^n|A|$; when n is odd this reads $|A| = -|A|$, forcing $|A| = 0$. No inverse — ever.
- **Row operations build the inverse too.** Row-reduce $[A \mid I]$ to $[I \mid A^{-1}]$; the syllabus focuses on the adjoint route, but knowing this one saves you in a pinch.

Now the payoff. A linear system $a_1x + b_1y = c_1$, $a_2x + b_2y = c_2$ is a matrix sentence: $AX = B$, with A holding the coefficients, $X = [x; y]$, and B the constants. If $|A| \neq 0$, multiply both sides on the left by A^{-1}: $A^{-1}AX = IX = X$, so $X = A^{-1}B$ — the **whole solution in one multiplication**. If $|A| = 0$, the method refuses politely: the system is either **inconsistent** (no solution, when $(\text{adj } A)B \neq O$) or has **infinitely many** solutions (when $(\text{adj } A)B = O$). That case-split, "unique / none / infinitely many", is precisely the "consistency, inconsistency and number of solutions" pointer.

EVERYDAY EXAMPLE — THE JUICER THAT UN-JUICES

A juicer turns fruit into juice, and no button turns juice back into fruit — but some machines genuinely run backwards: a washing machine that spins water out can be run to distribute it back, a lift goes up and then down. Multiplying by a non-singular matrix A is the reversible ride: it transforms X into B, and multiplying by A^{-1} runs the ride backwards, handing back X from B. A **singular** matrix is the juicer — information is flattened (many different X would give the same B), so there is no way back. The determinant is the ride's safety certificate: $|A| \neq 0$ means reversible; $|A| = 0$ means one-way.

WORKED EXAMPLE — SOLVING A SYSTEM BY THE MATRIX METHOD

Solve $3x + y = 5$, $5x + 2y = 9$ using matrices.

Solution. Step 1, write $AX = B$: $A = [3, 1; 5, 2]$, $X = [x; y]$, $B = [5; 9]$. Step 2, check the gate: $|A| = 3 \cdot 2 - 1 \cdot 5 = 1 \neq 0$, so a unique solution exists. Step 3, the 2×2 shortcut: $A^{-1} = \frac{1}{1}[2, -1; -5, 3] = [2, -1; -5, 3]$. Step 4, one multiplication:

$$X = A^{-1}B = [2, -1; -5, 3][5; 9] = [2 \cdot 5 + (-1) \cdot 9; -5 \cdot 5 + 3 \cdot 9] = [1; 2].$$

So $x = 1$, $y = 2$. Step 5, verify in the **original** equations: $3(1) + 2 = 5$ ✓ and $5(1) + 2(2) = 9$ ✓. Five steps, every one of them a mark.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — MATRICES AND DETERMINANTS

Formula	Letters mean	Fire it when
$A = [a_{ij}]_{m \times n}$	m rows, n columns; a_{ij} sits in row i, column j	locating or building entries
Equality: same order, every entry equal	position-by-position matching	"find x, y" questions
$(m \times n)(n \times p) = m \times p$	inner shapes must match	deciding if a product exists
Entry (i, j) of AB = row i of A · column j of B	dot product = sum of products	computing products
$(AB)' = B'A'$	order flips	transpose simplifications
$A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$	first part symmetric, second skew	sym + skew decomposition
$\left\lvert \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right\rvert = ad - bc$	cross-multiply and subtract	2×2 determinants

Formula	Letters mean	Fire it when
$C_{ij} = (-1)^{i+j} M_{ij}$	checkerboard signs	3×3 expansion
$\|kA\| = k^n \|A\|$	n = order of the matrix	scalar-multiple determinants
$\|A'\| = \|A\|$, $\|AB\| = \|A\|\|B\|$	transpose and product laws	determinant properties
Area $= \frac{1}{2}\|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\|$	absolute value; result 0 = collinear	triangle geometry
$A(\text{adj } A) = \|A\| I$, $\|\text{adj } A\| = \|A\|^{n-1}$	the adjoint identities	adjoint questions
$A^{-1} = \frac{\text{adj } A}{\|A\|}$, needs $\|A\| \neq 0$	singular gate	building inverses
$X = A^{-1}B$; if $\|A\| = 0$: none or infinitely many	A coefficients, B constants	matrix method

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — MATRICES AND DETERMINANTS

The mistake	What is right	The trap it sets
Multiplying without a shape check	$(m \times n)(n \times p)$ needs the inner numbers equal	Options list BA for shapes where only AB exists
Trusting $AB = BA$	Order matters; commutativity is the rare exception	"Which is always true?" questions with commutative-sounding lures
Cancelling A from $AB = AC$	Valid only if A^{-1} exists	Distractor "B = C" without the invertibility check
$\|kA\| = k\|A\|$	$\|kA\| = k^n\|A\|$ with n the order	The single most repeated determinant MCQ
Expanding $(A + B)^2 = A^2 + 2AB + B^2$	Honest expansion: $A^2 + AB + BA + B^2$	The fake "algebra-style" option sits right there
Skew-symmetric with a non-zero diagonal	Every diagonal entry of a skew matrix is 0	One-row type-identification questions
Writing $(AB)^{-1} = A^{-1}B^{-1}$	$(AB)^{-1} = B^{-1}A^{-1}$ — order flips	Mirror-image inverse options

The mistake	What is right	The trap it sets
Finding an inverse for $\lvert A\rvert = 0$	Singular means no inverse exists	Options include a computed "inverse" that cannot exist
Quoting a negative triangle area	Area is the absolute value, always ≥ 0	Negative-value distractors in area questions

Sixty-Second Self-Test

Judge each statement fast — say true or false out loud before checking the verdicts.

Statement	Verdict
A 2×3 matrix has a determinant.	False — determinants live only on square matrices.
Every diagonal matrix is symmetric.	True — a diagonal matrix is its own transpose.
If $AB = O$, then $A = O$ or $B = O$.	False — zero divisors exist for matrices.
For a 3×3 matrix, $\lvert A'\rvert = -\lvert A\rvert$.	False — transposing never flips the sign.
If $\lvert A\rvert = 0$, the system $AX = B$ has no solution.	False — it has either none or infinitely many.
$A + A'$ is symmetric for every square A.	True — transpose it and watch it return.

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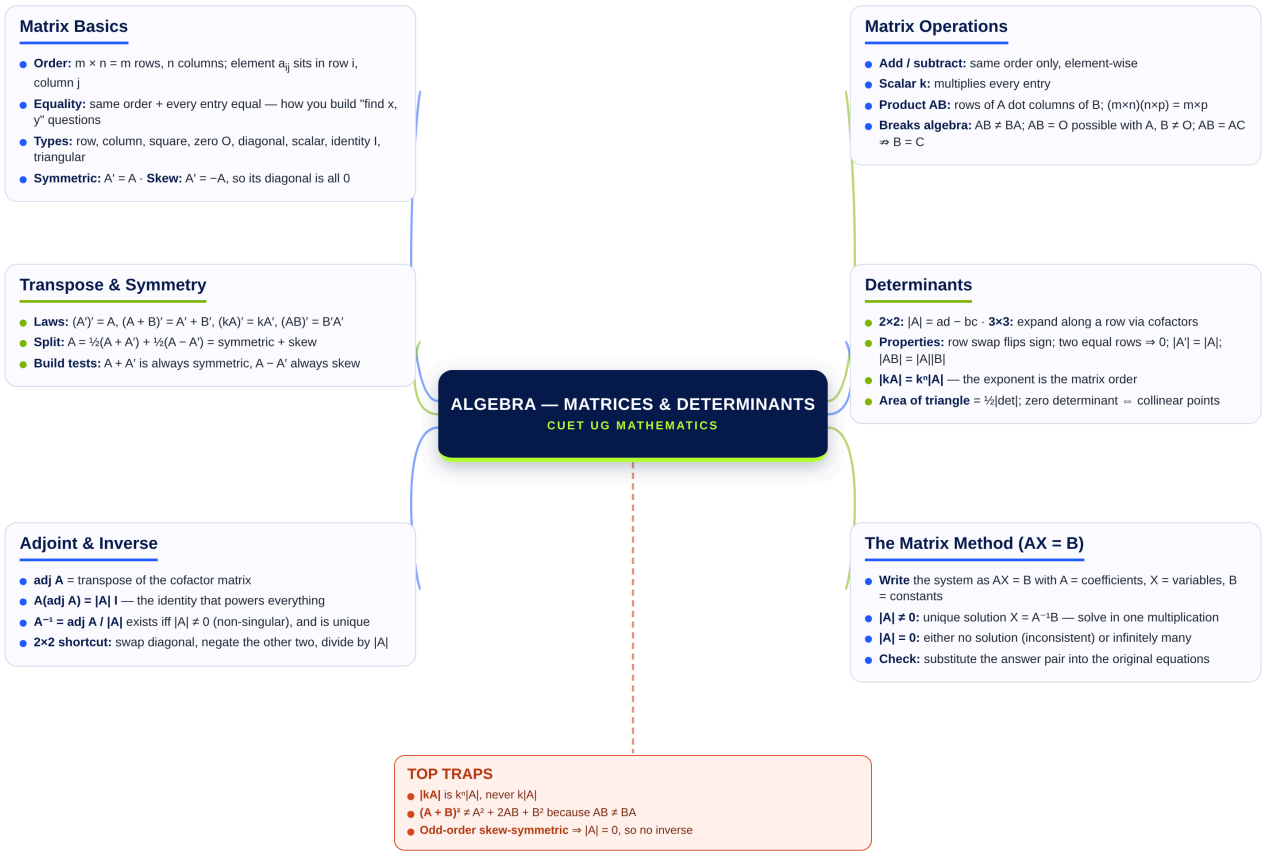


Figure 2: The whole chapter in one look: six branches from matrix basics to the matrix method, with the traps that sit underneath them.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1, so attempt from understanding, not hope. Every question below is answerable in under ninety seconds with a technique from this chapter — if you are writing a page of algebra, the shortcut is hiding.

Q1. For $A = [2, -1; 3, 4]$, which statement is true?

(A) A is symmetric
(B) A is skew-symmetric
(C) A is a scalar matrix
(D) A is neither symmetric nor skew-symmetric

Q2. If $[x - y, 4; 2, x + y] = [3, 4; 2, 9]$, then (x, y) equals:

(A) $(6, 3)$
(B) $(3, 6)$
(C) $(9, 3)$
(D) $(6, -3)$

Q3. For $A = [1, 2; 3, 4]$ and $B = [0, 1; 1, 0]$, the matrix $AB - BA$ equals:

(A) $[1, 3; -3, -1]$ **(B)** $[-1, -3; 3, 1]$
(C) $[2, 1; 4, 3]$ **(D)** $[0, 0; 0, 0]$

Q4. For $A = [1, 1; 2, 2]$ and $B = [1, 1; -1, -1]$, the product $AB = O$ even though neither matrix is O. This shows that:

(A) multiplication of matrices is commutative **(B)** $AB = O$ always forces $A = O$ or $B = O$
(C) matrices have no zero divisors **(D)** $AB = O$ need not force $A = O$ or $B = O$

Q5. If $A = [3, -2; 4, -2]$, then $|A|$ equals:

(A) -2 **(B)** -14
(C) 2 **(D)** 14

Q6. If A is a 3×3 matrix with $|A| = 4$, then $|2A|$ equals:

(A) 8 **(B)** 24
(C) 32 **(D)** 64

Q7. The area of the triangle with vertices $(0, 0)$, $(4, 0)$, $(0, 3)$ is:

(A) 7 square units **(B)** 12 square units
(C) 3.5 square units **(D)** 6 square units

Q8. For $A = [2, 3; 1, 4]$, the matrix A^{-1} is:

(A) $[4, 3; 1, 2]$ **(B)** $\left[\frac{4}{5}, -\frac{3}{5}; -\frac{1}{5}, \frac{2}{5}\right]$
(C) $[4, -3; -1, 2]$ **(D)** $\left[\frac{2}{5}, -\frac{1}{5}; -\frac{3}{5}, \frac{4}{5}\right]$

Q9. The system $x + 2y = 3$, $2x + 4y = 7$ has:

(A) a unique solution **(B)** no solution
(C) infinitely many solutions **(D)** exactly two solutions

Q10. If A is a skew-symmetric matrix of order 3, then $|A|$ equals:

(A) 0 **(B)** 1
(C) -1 **(D)** depends on the entries of A

Detailed Solutions

Sol. Q1. $A' = [2, 3; -1, 4]$ by swapping rows and columns. Compare: $A' \neq A$ (not symmetric) and $A' \neq -A$ (skew would need $[2, 3; -1, 4] = [-2, 1; -3, -4]$). Quick dead-giveaway: a skew matrix must have zeros on its diagonal, but $a_{11} = 2$.

Answer: (D)

Sol. Q2. Match entries: $x - y = 3$ and $x + y = 9$. Adding: $2x = 12$, so $x = 6$; subtracting: $2y = 6$, so $y = 3$. The middle entries (4 and 2) are already equal — they exist to confirm the matrices really can match.

Answer: (A)

Sol. Q3. $AB = [1 \cdot 0 + 2 \cdot 1, 1 \cdot 1 + 2 \cdot 0; 3 \cdot 0 + 4 \cdot 1, 3 \cdot 1 + 4 \cdot 0] = [2, 1; 4, 3]$ and $BA = [3, 4; 1, 2]$. Subtracting entry-wise: $AB - BA = [-1, -3; 3, 1]$. This question **is** the non-commutativity lesson with numbers attached.

Answer: (B)

Sol. Q4. Both matrices are non-zero, yet every dot product is $1 - 1 = 0$, so $AB = O$. Matrices therefore have **zero divisors** — the number property "product zero forces a factor zero" fails here. Nothing about commutativity is involved.

Answer: (D)

Sol. Q5. Cross-multiply: $|A| = 3 \cdot (-2) - (-2) \cdot 4 = -6 + 8 = 2$. Watch the double negative — dropping the minus on $(-2)(4)$ is exactly how the -14 distractor is built.

Answer: (C)

Sol. Q6. Every entry of $2A$ is 2 times an entry of A. Each of the 3 rows contributes a factor of 2, so $|2A| = 2^3|A| = 8 \cdot 4 = 32$. The exponent is the order — here 3, not 2.

Answer: (C)

Sol. Q7. Determinant route: $\frac{1}{2}|0(0-3) + 4(3-0) + 0(0-0)| = \frac{1}{2}|12| = 6$. Or notice this is a right triangle with legs 4 and 3: $\frac{1}{2} \cdot 4 \cdot 3 = 6$ — the determinant agrees with the geometry.

Answer: (D)

Sol. Q8. Gate: $|A| = 2 \cdot 4 - 3 \cdot 1 = 5 \neq 0$. Shortcut — swap the diagonal, negate the other two, divide by 5: $A^{-1} = \left[\frac{4}{5}, -\frac{3}{5}; -\frac{1}{5}, \frac{2}{5}\right]$. Option (C) is the adjoint without the division; option (D) has swapped-off-diagonal signs.

Answer: (B)

Sol. Q9. Here $|A| = |[1, 2; 2, 4]| = 4 - 4 = 0$, so the matrix method stalls at the gate — no unique solution. Check which side: adj $A = [4, -2; -2, 1]$, then $(\text{adj } A)B = [4 \cdot 3 - 2 \cdot 7; -2 \cdot 3 + 1 \cdot 7] = [-2; 1] \neq O$ — inconsistent, no solution. The equations are the same line doubled up… almost: $x + 2y = 3$ vs $x + 2y = 3.5$ after dividing the second by 2. Parallel lines never meet.

Answer: (B)

Sol. Q10. For skew-symmetric A: $A' = -A$, so $|A| = |A'| = |-A| = (-1)^3|A| = -|A|$, which forces $|A| = 0$ — for **every** 3×3 skew-symmetric matrix, no matter the entries.

Answer: (A)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(A) in Q1 — "symmetric"	Checked $A' = A$ but never $A' = -A$	Two-check ritual: test both, and use the diagonal-zero test for skew first
(C) in Q3 — "the product itself"	Computed AB and forgot the subtraction	Read the expression aloud before computing: "AB minus BA"
(B) in Q6 — "24"	Treated $\|2A\|$ as $2 \cdot \|A\|$ plus… nothing consistent	$\|kA\| = k^n\|A\|$ — write the power before touching numbers
(C) in Q8 — "the adjoint"	Built adj A but skipped dividing by $\|A\|$	Inverse is a two-step product: adjoint, then scale by $\frac{1}{\|A\|}$
(A) in Q9 — "unique"	Answered from the shape of the equations, not $\|A\|$	Determinant first, conclusions second — always
(D) in Q10 — "depends"	Tried to imagine a special non-singular skew matrix	The $(-1)^n$ argument closes every odd-order skew case at once

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